

Final Project Proposal: Visualizing Global Hotel Booking Cancellation Patterns

Group Members: KitKi Chim, Chris Huang, Yunxiao Li

Project Overview

This project develops an interactive data visualization dashboard to explore hotel booking cancellation behavior. Rather than focusing on prediction, our goal is to make cancellation patterns visually interpretable—helping users understand when, where, and under what booking conditions cancellations occur.

The final output will be a publicly accessible interactive web application (Streamlit) that enables users to explore cancellation behavior across time, booking structures, and geographic origin.

Planned Visualizations and Analytical Logic

1. Geospatial Map — Cancellation Rate by Guest Origin Country

We will build a choropleth world map showing where cancellations come from. Each country will be shaded by its average cancellation rate, and users will be able to filter by hotel location (e.g., Tokyo city hotel vs. Sydney resort hotel) and by season. This will reveal which guest origins — such as domestic vs. international — are associated with higher or lower cancellation rates, and whether patterns differ across hotel markets.

2. Network Graph — Booking Channel × Market Segment Relationships

We will construct a bipartite network graph where one set of nodes represents booking channels (e.g., Online TA, Direct, Corporate, GDS) and the other set represents market segments (e.g., Transient, Group, Contract). Edge weight encodes cancellation volume, and edge color encodes average cancellation rate. This reveals which channel–segment combinations are most cancellation-prone and how the booking ecosystem is structured differently across hotel locations.

3. Temporal Heatmap — Lead Time vs. Arrival Month Cancellation Risk

A 2D heatmap with lead time bins on one axis and arrival month on the other, colored by cancellation rate. This visualization makes it immediately clear whether booking far in advance is risky across all seasons, or only for certain months (e.g., peak travel seasons in Japan during cherry blossom or Golden Week). Users can toggle between hotel locations and deposit type to explore interaction effects.

4. (Supporting) Parallel Coordinates Plot — Multi-variable Cancellation Profiling

A parallel coordinates chart allowing users to trace individual booking profiles across key variables: lead time, average daily rate (ADR), length of stay, number of special requests, and cancellation outcome. This supports exploratory comparison of "typical cancelled" vs. "typical kept" reservation profiles.

Data Sources

Source	Description	Access
Hotel Booking Demand Dataset (Antonio et al., 2019)	119,210 reservations from two Portuguese hotels — used as a baseline reference and for method validation	Kaggle / ScienceDirect open access
Booking.com Public API / Web data	Current hotel property metadata (location, type, star rating) to enrich geospatial layers	API / scraping (subject to terms)
Expedia Partner Solutions Open Data	Hotel performance and booking channel data for Asia-Pacific properties	EPS open data portal
UNWTO Tourism Statistics	International tourist arrival counts by country of origin — used to normalize cancellation rates by travel volume	UNWTO Data
OpenStreetMap / Mapbox GL	Base map tiles for geospatial visualization	Free / Mapbox free tier

(All datasets are either publicly available, open-licensed, or accessible via free API tiers. We will document exact download links and licenses in the project repository.)

Specialized Visualization Types and Tools

The project will incorporate two required specialized visualization types: geospatial mapping and network visualization. These will be implemented using GeoPandas and Plotly for spatial analysis, and NetworkX with PyVis for network construction and rendering.

In addition, we will use Pandas for data processing, Plotly or Altair for general visualization, and Streamlit to build and deploy the interactive web application.

Potential Hurdles

A key challenge lies in balancing complexity with clarity. Network visualizations, in particular, can quickly become cluttered, requiring careful filtering and design choices to maintain readability. Similarly, choropleth maps can be misleading if not properly normalized, especially when comparing countries of vastly different sizes.

Another challenge is methodological: choices such as how to bin lead time or aggregate data can significantly influence the patterns observed in the visualizations. We will therefore test multiple specifications to ensure robustness.

Finally, integrating multiple views into a coherent and responsive dashboard introduces technical complexity. To address this, we will prioritize the most informative views and implement interaction features incrementally.

Expected Outcome

The final product will be a polished, interactive visualization platform that allows users to explore hotel booking cancellations from multiple perspectives. By combining spatial, temporal, and network-based analysis, the project aims to demonstrate both technical proficiency and a strong understanding of visualization principles, particularly in terms of clarity, interpretability, and user interaction.